

Polynomial Functions Quiz

Algebra 2 Version A

Name: _____

1. (4 pts) Factor completely: $2x^2 - 2x - 12$

Answer: $2(x - 3)(x + 2)$

2. (8 pts) Let $f(x) = x^2 + 3x - 4$. Use algebra to answer the following.

a. Find all real zeros of f .

$-4, 1$

b. Find the y -intercept.

$(0, -4)$

c. Describe the end behavior of f .

$f(x) \rightarrow +\infty$ as $x \rightarrow \pm\infty$

d. State the interval(s) where $f(x) < 0$.

$(-4, 1)$

3. (10 pts) Let $f(x) = x^3 - 5x^2 + 2x + 8$. Use algebra to answer the following.

a. Find all real zeros of f .

$-1, 2, 4$

b. Find the y -intercept.

$(0, 8)$

c. Describe the end behavior of f .

$f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$, $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$

d. State the interval(s) where $f(x) > 0$.

$(-1, 2) \cup (4, +\infty)$

e. What is the maximum number of turning points f can have?

2